

SUPPORTLYTICS

Optimizing IT Support Team Performance Using Analytics

Final Project Report

Project Domain: IT Support Analytics

Technology Stack: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI

Dataset Source: Internal IT Support Ticket System

Project Type: Data Analytics, Visualization and Performance Optimization

Submitted By

Balu

1. Executive Summary

Supportlytics is an end-to-end analytics project designed to evaluate IT support ticket operations and identify opportunities to improve service efficiency. The project uses ticket attributes such as type, priority, category, country, queue, resolution duration, cluster information and similarity scores to understand workload patterns and support performance.

The analysis follows a complete data analytics workflow: data acquisition and understanding, data cleaning, feature engineering, exploratory analysis, cluster and similarity analysis, performance evaluation, geographic analysis, dashboard development and final reporting.

The final outcome is a performance-oriented analytical solution that converts raw support ticket records into visual insights and actionable recommendations. The dashboard and visualizations help stakeholders monitor resolution efficiency, ticket concentration, issue clusters, priority workload and regional performance.

2. Project Objectives

- Understand and preprocess the IT support ticket dataset for reliable analysis.
- Measure ticket volume, priority distribution, issue categories and ticket types.
- Calculate resolution duration and other performance-oriented features.
- Identify frequently occurring issue clusters and analyze their similarity scores.
- Compare resolution performance across priorities, countries, teams and categories.
- Identify high-priority unresolved issues and potential performance gaps.
- Create meaningful visualizations and an interactive performance dashboard.
- Provide actionable recommendations for workflow improvement and resource allocation.

3. Problem Statement

IT support teams handle large numbers of requests, incidents and problems across different categories, priorities and geographic regions. Without structured analysis, it can be difficult to identify workload concentration, slow-resolution areas, recurring technical issues and resource bottlenecks.

Supportlytics addresses this challenge by analyzing historical support ticket data to uncover performance trends and recurring patterns. The project aims to support data-driven decisions that can reduce resolution time, improve prioritization and increase overall service efficiency.

4. Project Scope

- Ticket-level data preparation and quality validation.
- Descriptive and exploratory analysis of support operations.
- Similarity and cluster-level issue analysis.
- Performance comparison by priority, type, category, country and queue.
- Geographic visualization of ticket concentration and issue categories.
- Dashboard creation using Power BI and supporting Python visualizations.
- Documentation of insights and recommendations.

5. Dataset Understanding

The project uses an internal IT Support Ticket System dataset. Each record represents a support ticket and contains operational information that can be used to evaluate ticket demand and resolution performance.

Data Area	Representative Fields	Purpose
Ticket Information	Ticket ID, Date, Type	Identify and classify support records
Priority	Priority, Priority Score	Measure urgency and workload
Issue Classification	Category, Cluster ID, Cluster Name	Identify recurring issue groups
Performance	Resolution Date, Resolution Duration	Measure support efficiency
Similarity	Similarity Score, Similarity Level	Measure issue similarity
Geography	Country, Latitude, Longitude	Analyze regional patterns
Operations	Queue / Team	Compare workload and handling performance

6. Key Performance Indicators (KPIs)

KPI	Definition	Business Use
Average Resolution Time	Average time taken to resolve tickets	Monitor service efficiency
Ticket Volume	Number of support tickets	Measure workload and demand
Priority Distribution	Tickets grouped by priority	Understand urgency and staffing needs
Resolution Rate	Resolved tickets relative to total tickets	Evaluate operational effectiveness
Cluster Similarity Index	Average similarity score of issue groups	Identify recurring / related issues
Top Categories	Most frequent issue categories	Prioritize recurring technical problems
Regional Performance	Ticket and resolution metrics by country/region	Identify geographic performance gaps

7. Methodology

1. Load the raw CSV dataset using Pandas.
2. Inspect schema, data types, duplicate records and missing values.
3. Clean incomplete or inconsistent text and date fields.
4. Convert date columns to appropriate datetime formats.
5. Create Resolution_Duration / Resolution_Time from resolution and ticket dates.
6. Create Priority_Score and categorize Similarity_Score into meaningful performance buckets.
7. Encode categorical fields where required for analytical or clustering workflows.
8. Perform exploratory analysis using Python visualization libraries.
9. Analyze issue clusters, similarity scores and cluster-level resolution performance.
10. Compare resolution performance by priority, type, category, country and queue.
11. Build geographic and comparative visualizations.
12. Combine the most important metrics into an interactive Power BI dashboard.
13. Interpret the findings and translate them into operational recommendations.

8. Milestone-wise Implementation

Milestone 1 – Data Readiness — Dataset loading, schema inspection, missing-value handling, cleaning and feature engineering.

Milestone 2 – Visualization & Clustering — EDA, ticket distributions, category/cluster analysis, similarity analysis and performance-gap visualizations.

Milestone 3 – Performance & Geography — Resolution-time analysis, high-priority issue identification, regional analysis and geographic visualizations.

Milestone 4 – Dashboard & Documentation — Power BI dashboard, KPI layout, final report, presentation and GitHub documentation.

9. Data Cleaning and Feature Engineering

Data preparation was performed before analysis to ensure that the results were based on consistent and usable records. Missing, incomplete and incorrectly formatted fields were reviewed and handled according to their analytical importance.

- Checked column names, data types and structural consistency.
- Reviewed missing and incomplete values.
- Removed or corrected invalid date and text values where applicable.
- Converted date fields to datetime format.
- Created Resolution_Duration / Resolution_Time using the difference between resolution date and ticket date.
- Created Priority_Score for quantitative priority analysis.
- Grouped Similarity_Score into interpretable Similarity_Level buckets.
- Encoded categorical variables for analytical and clustering requirements.
- Saved a processed dataset for downstream analysis.

10. Exploratory Data Analysis

EDA was used to understand the distribution of support tickets and identify the areas that required deeper investigation.

- Ticket type distribution across Request, Problem and Incident.
- Ticket volume by country and priority.

- Most frequent issue categories such as Integration, Security and Bug.
- Ticket distribution across support queues.
- Frequency of issue clusters and cluster names.
- Resolution-duration patterns across ticket groups.
- Comparison of high-, medium- and low-priority workload.

11. Cluster and Similarity Analysis

Cluster and similarity analysis was used to determine whether support tickets were concentrated around recurring issue groups. Cluster_ID and Cluster_Name were analyzed together with Similarity_Score to identify common technical patterns.

- Calculated and compared average similarity scores within clusters.
- Analyzed cluster size to identify frequently occurring issue groups.
- Compared cluster composition with ticket type and category.
- Compared similarity levels with resolution performance.
- Used scatter plots and boxplots to identify performance gaps and outliers.
- Highlighted recurring clusters that may benefit from standard troubleshooting procedures or automation.

12. Performance Trend Analysis

Performance analysis focused on resolution efficiency and operational differences between ticket groups. Average resolution duration was compared across priority, ticket type, country, category and support queue.

- Compared average resolution times by priority.
- Compared resolution times across Request, Problem and Incident tickets.
- Identified countries or teams with comparatively faster handling performance.
- Reviewed unresolved high-priority tickets as operational risks.
- Identified categories and clusters with longer resolution durations.
- Used comparative charts to identify potential staffing and process gaps.

13. Geographic and Category-Level Analysis

Geographic analysis was used to understand where support demand was concentrated and whether regional differences were visible in ticket volume or resolution performance.

- Created regional ticket concentration heatmaps.
- Mapped issue categories using latitude and longitude where available.
- Compared ticket volume across countries.
- Compared regional resolution performance.
- Evaluated the relationship between cluster size and performance score.
- Used geographic patterns to support resource allocation recommendations.

14. Visualization Portfolio

Visualization	Purpose	Recommended Interpretation
Ticket Type Bar Chart	Show Request / Problem / Incident distribution	Identify dominant ticket types
Priority Distribution Chart	Compare workload by priority	Identify urgency concentration
Top Category Chart	Rank issue categories	Find recurring technical issues
Country Distribution Chart	Compare ticket volume geographically	Identify high-demand regions
Resolution Time by Priority	Compare average resolution duration	Detect priority-related performance gaps
Cluster Frequency Chart	Rank clusters by ticket count	Identify recurring issue groups
Similarity Boxplot	Compare similarity-score distributions	Detect clusters with unusual similarity
Cluster vs Performance Scatter Plot	Compare cluster size/similarity with performance	Identify high-volume problem areas
Regional Heatmap	Show geographic ticket concentration	Identify support-demand hotspots
Geographic Issue Map	Plot issue categories geographically	Understand regional issue patterns

Milestone 1: Week 1–2

Module 1: Project Initialization & Dataset Setup

- Defined project objectives, KPIs and analysis workflow.
- Loaded the IT support ticket dataset using Pandas.
- Checked dataset structure, data types, missing values and duplicates.
- Analyzed initial ticket distribution by Type, Priority and Category.

	count	unique		top	freq	mean	std	min	25%	50%	75%	max
Ticket_ID	1000000.0	NaN		NaN	NaN	500000.5	288675.278932	1.0	250000.75	500000.5	750000.25	1000000.0
Date_Created	1000000	30		2023-11-03	33717	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Date_Resolved	700057	36		2023-11-21	23507	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Category	1000000	5		Software	200628	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Subcategory	1000000	493	Laptop touch screen not working		2166	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Priority	1000000	4		Low	250419	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Status	1000000	5		Resolved	200903	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Assigned_Team	1000000	4		Desktop Support	399425	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Resolution_Time_Hrs	1000000.0	NaN		NaN	NaN	5.246152	2.742932	0.5	2.87	5.24	7.62	10.0
Description	1000000	493	User reported Laptop touch screen not working....		2166	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Escalated	1000000	2		False	500089	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
<class 'pandas.DataFrame'>
RangeIndex: 1000000 entries, 0 to 999999
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Ticket_ID             1000000 non-null int64
1   Date_Created          1000000 non-null str
2   Date_Resolved         700057 non-null str
3   Category              1000000 non-null str
4   Subcategory           1000000 non-null str
5   Priority               1000000 non-null str
6   Status                1000000 non-null str
7   Assigned_Team         1000000 non-null str
8   Resolution_Time_Hrs   1000000 non-null float64
9   Description            1000000 non-null str
10  Escalated              1000000 non-null bool
dtypes: bool(1), float64(1), int64(1), str(8)
memory usage: 77.2 MB
```

Module 2: Data Cleaning & Feature Engineering

- Cleaned missing and inconsistent data.
- Converted date columns into proper datetime format.
- Created Resolution_Duration to measure ticket resolution time.
- Created Priority_Score for priority-based analysis.
- Prepared and saved the cleaned dataset.

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.metrics.pairwise import cosine_similarity

# Cluster on the 493 unique subcategories only
unique_subcats = df['Subcategory'].drop_duplicates().reset_index(drop=True)
print(f"Unique subcategories to cluster: {len(unique_subcats)}")

tfidf = TfidfVectorizer(stop_words='english', max_features=300)
tfidf_matrix = tfidf.fit_transform(unique_subcats)

N_CLUSTERS = 15
kmeans = KMeans(n_clusters=N_CLUSTERS, random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(tfidf_matrix)

# Similarity of each subcategory to its own cluster centroid
centroids = kmeans.cluster_centers_
similarity_scores = [
    cosine_similarity(tfidf_matrix[i], centroids[cluster_labels[i]].reshape(1, -1))[0][0]
    for i in range(len(unique_subcats))
]

cluster_map = pd.DataFrame({
    'Subcategory': unique_subcats,
    'Cluster_ID': cluster_labels,
    'Similarity_Score': similarity_scores
})
cluster_map.head(10)
```

Unique subcategories to cluster: 493

	Subcategory	Cluster_ID	Similarity_Score
0	Access denied by auditor policy	10	0.973886
1	Suspicious DLL injection	0	0.464938
2	Suspicious SOC alert	0	0.455810
3	App freezing	1	0.084434
4	Access logging failure	9	0.481351
5	Suspicious outbound traffic	0	0.508304
6	Unauthorized access attempt	1	0.307032

Is_Resolved	False	True
Status		
In Progress	59593	139842
On Hold	59951	140568
Open	60044	139758
Pending	60031	139310
Resolved	60324	140579

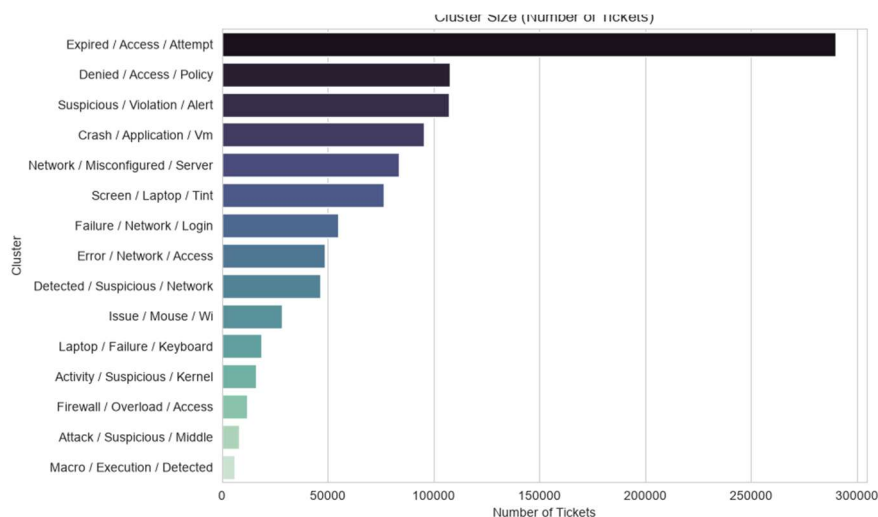
Milestone 2: Week 3–4

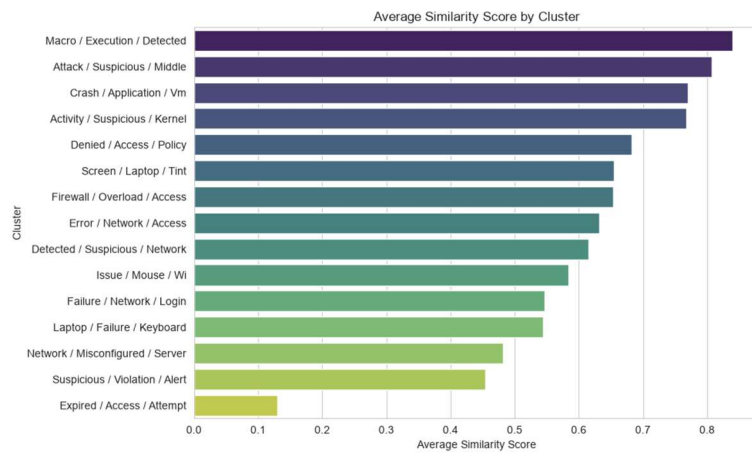
Module 3: Exploratory Visualization

- Analyzed ticket distribution by Type, Priority, Category and Queue.
- Identified the most frequent categories and issue clusters.
- Compared ticket volumes across different groups.
- Created multiple charts to identify important patterns.

Module 4: Similarity & Cluster Insights

- Analyzed similarity scores within issue clusters.
- Compared cluster size with ticket types and categories.
- Used boxplots and scatter plots to identify performance gaps.
- Identified recurring and high-volume issue clusters.

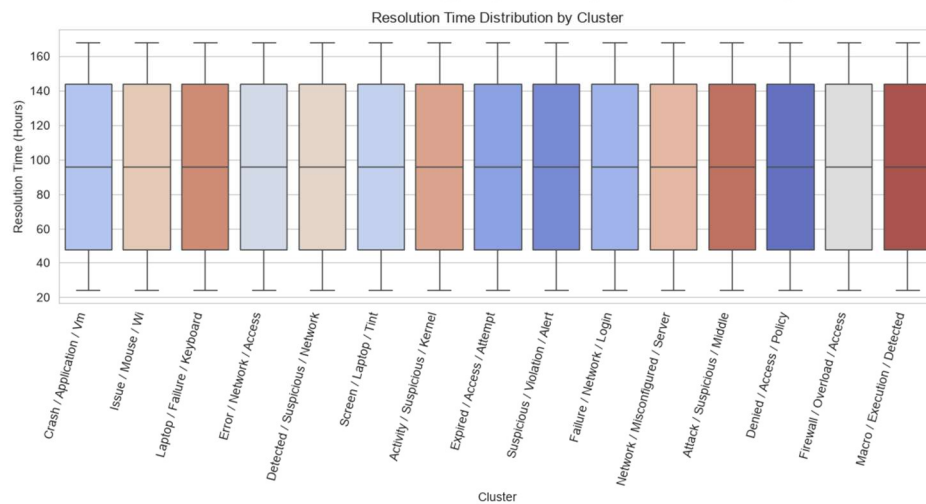




Milestone 3: Week 5–6

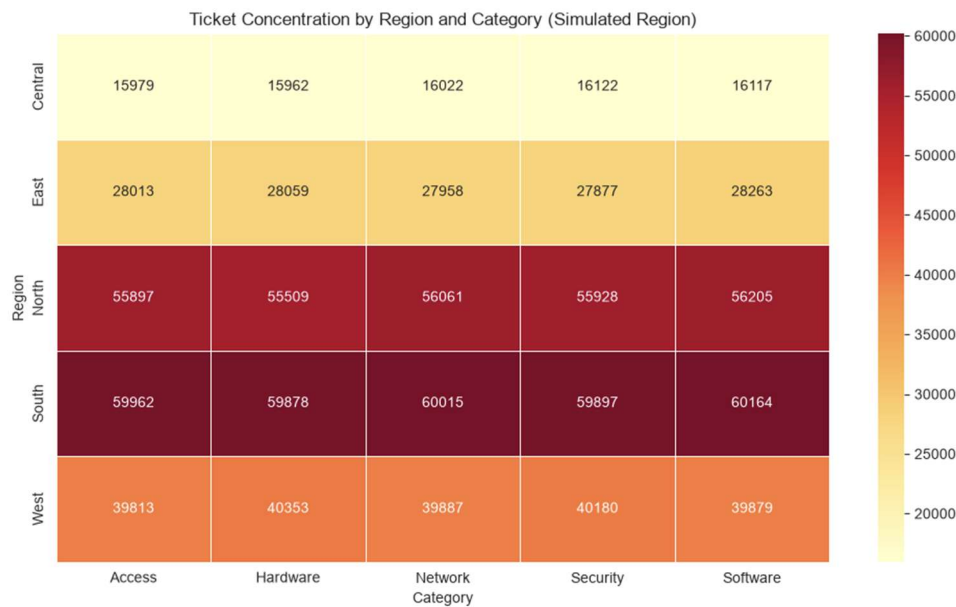
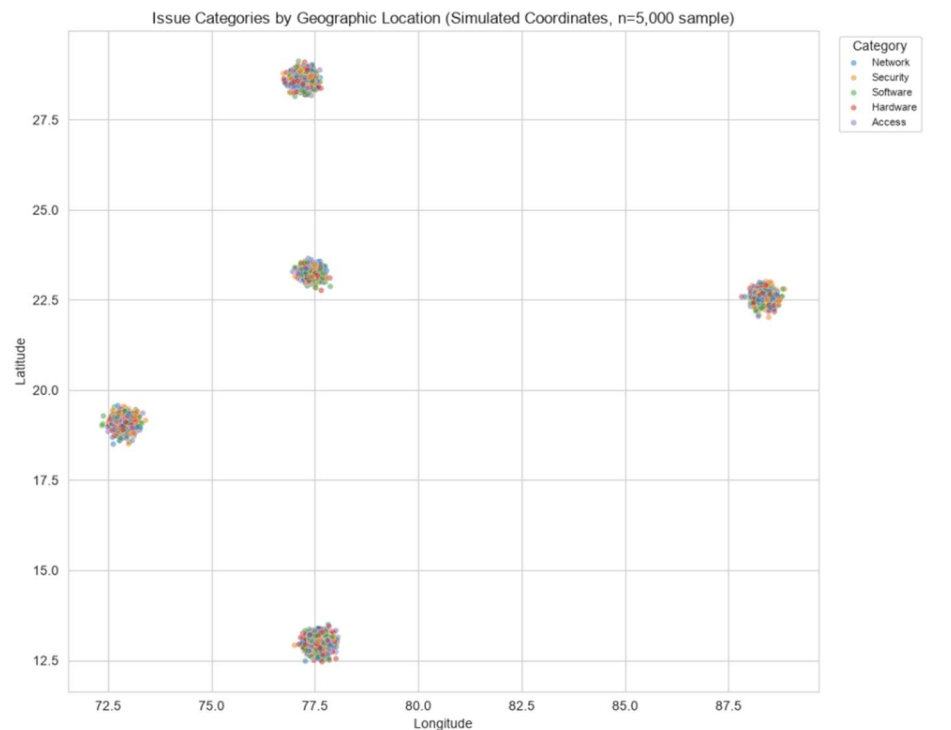
Module 5: Performance Trend Analysis

- Compared average resolution time by Priority and Ticket Type.
- Analyzed resolution performance across support teams/queues.
- Identified faster and slower-performing teams.
- Identified unresolved high-priority tickets and potential risks.



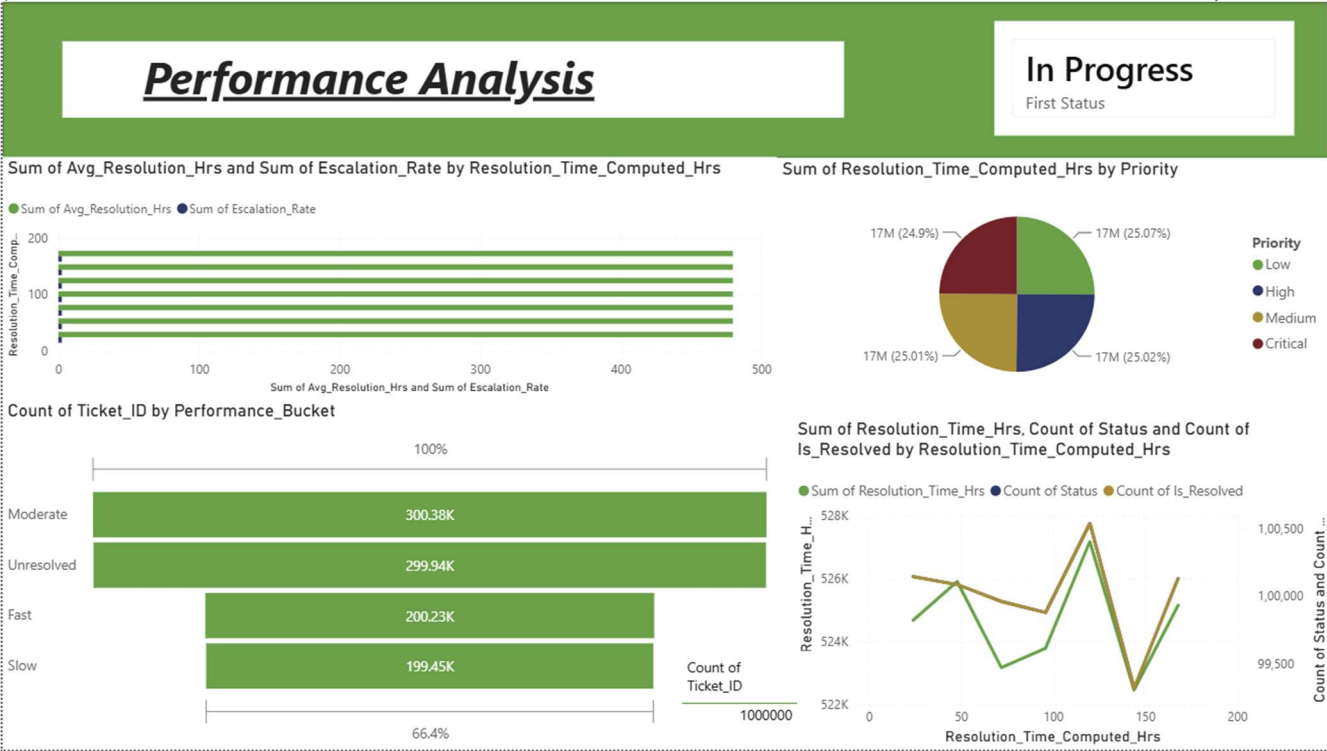
Module 6: Geographic & Category-Level Insights

- Analyzed ticket concentration across regions.
- Created geographic visualizations for ticket and category distribution.
- Compared regional performance.
- Analyzed the relationship between cluster size and performance.



15. Power BI Dashboard

The final dashboard consolidates the project findings into an interactive management view. The dashboard is designed to allow users to monitor operational KPIs and drill into priority, category, cluster and geographic dimensions.



Geography Analysis

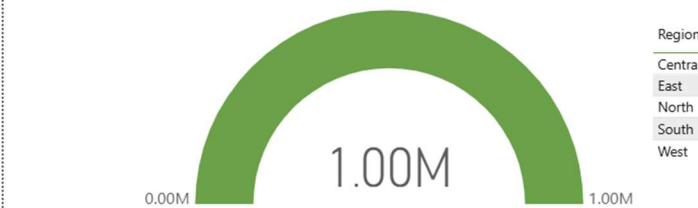
67M

Sum of Resolution_Time_C...

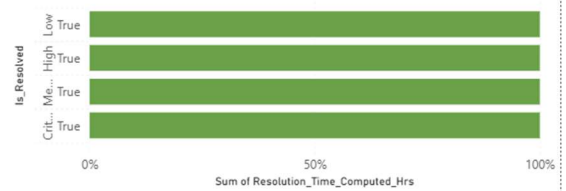
5

Count of Region

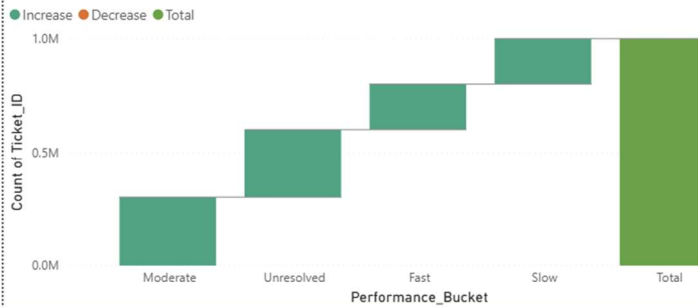
Count of Category and Count of Is_Resolved



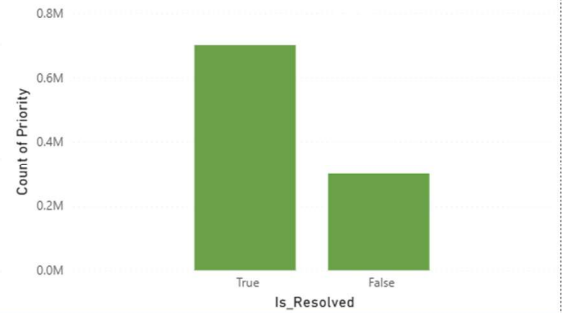
Sum of Resolution_Time_Computed_Hrs by Priority and Is_Resolved



Count of Ticket_ID by Performance_Bucket



Count of Priority by Is_Resolved



Cluster Analysis

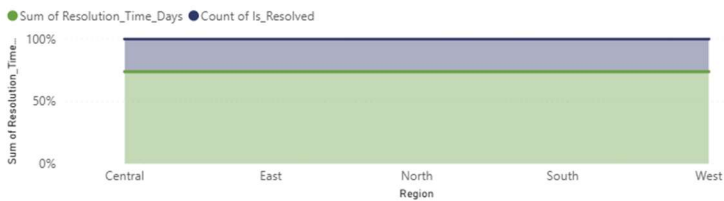
Access

Hardware

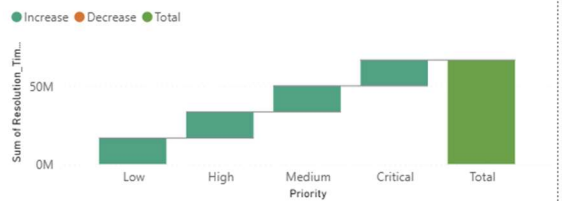
67M

Sum of Resolution_Time_Computed_Hrs

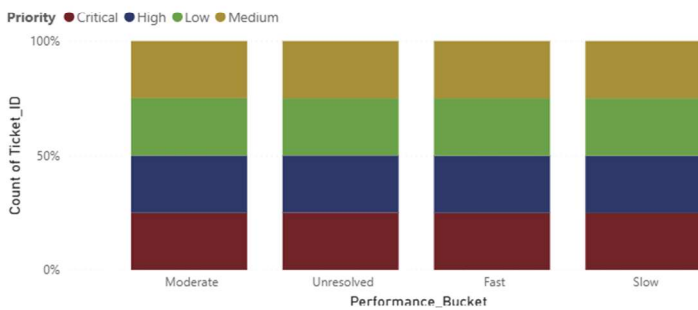
Sum of Resolution_Time_Days and Count of Is_Resolved by Region



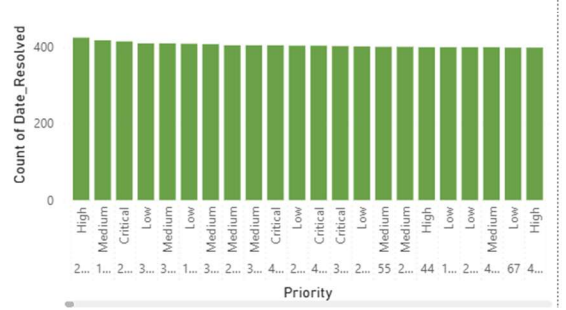
Sum of Resolution_Time_Computed_Hrs by Priority



Count of Ticket_ID by Performance_Bucket and Priority



Count of Date_Resolved by Subcategory_Encoded and Priority





- KPI Card – Average Resolution Time
- KPI Card – Total Tickets
- KPI Card – Resolution Rate
- KPI Card – Cluster Similarity Index
- KPI Card – Top Performing Region

- Bar Chart – Most Frequent Categories
- Chart – Ticket Distribution by Priority and Type
- Chart – Resolution Efficiency by Priority
- Chart – Cluster Performance vs Similarity Score
- Map – Ticket and Category Distribution by Region
- Interactive filters for country, priority, type, category, queue and cluster.

16. Key Analytical Insights

The completed analysis is intended to convert descriptive patterns into operational insights. The exact numerical values should be taken directly from the final dataset and dashboard rather than being assumed in this report.

Insight Area	Finding to Report from Final Dashboard
Ticket Demand	State the highest-volume ticket type, category and region.
Priority	State which priority contributes the largest workload and whether high-priority tickets are being resolved efficiently.
Resolution	State the average resolution time and identify the groups with the longest resolution duration.
Clusters	State the largest recurring clusters and their average similarity / resolution performance.
Geography	State the highest-demand regions and any meaningful regional performance differences.
Unresolved Risk	State the number or proportion of unresolved high-priority issues shown in the completed analysis.

17. Recommendations

14. Prioritize resource allocation toward categories and clusters with high ticket volume and long resolution durations.
15. Create standardized troubleshooting playbooks for recurring issue clusters.

16. Introduce stronger escalation rules for unresolved high-priority tickets.
17. Monitor resolution duration by queue/team to identify training or staffing requirements.
18. Use cluster and similarity analysis to identify opportunities for ticket auto-routing and knowledge-base recommendations.
19. Review regional workload regularly and rebalance support capacity when sustained demand differences are observed.
20. Track dashboard KPIs continuously rather than relying only on periodic reports.
21. Use historical ticket patterns to support workforce planning and preventive technical improvements.

18. Technology Stack

Technology	Role in Project
Python	Primary programming and analytics environment
Pandas	Data loading, cleaning, transformation and aggregation
NumPy	Numerical operations and feature calculations
Matplotlib	Analytical visualizations
Seaborn	Statistical and comparative visualizations
Plotly	Interactive visualizations
Power BI	Interactive dashboard and KPI reporting
Jupyter Notebook	Analysis documentation and experimentation
GitHub	Source-code and project documentation management

19. Project Deliverables

- Cleaned and processed IT support ticket dataset.
- Feature engineering and data dictionary documentation.
- Jupyter Notebook / Python analysis files.
- Minimum eight analytical visualizations and supporting charts.

- Cluster and similarity analysis.
- Performance metrics and geographic analysis.
- Interactive Power BI dashboard.
- Final analytical report.
- Presentation slides.
- GitHub repository containing project code, visuals and documentation.

20. Conclusion

Supportlytics demonstrates how data analytics can be applied to IT support operations to move from raw ticket records to measurable performance insights. By combining data cleaning, feature engineering, exploratory analysis, cluster and similarity analysis, geographic visualization and dashboard reporting, the project provides a structured view of support workload and efficiency.

The solution can help support managers identify recurring technical issues, monitor resolution performance, recognize high-demand regions and prioritize operational improvements. The most important next step is to continue tracking the dashboard KPIs over time so that recommendations can be validated using future ticket data.

Dashboard Link:

<https://drive.google.com/file/d/18Lhv7gliv2hVgEDg5u8RL4jISgSYRSUu/view?usp=sharing>

CSV File Link:

https://drive.google.com/file/d/1sW7eXfe29a_bpz3svSOjc5z1SlTHHoln/view?usp=sharing

21. Future Scope

- Add automated daily or weekly dashboard refresh.
- Develop predictive models for ticket resolution time.
- Predict escalation risk for high-priority tickets.
- Implement intelligent ticket routing using historical clusters.
- Use NLP on ticket descriptions for automated category and cluster assignment.

22. Appendix – Data Dictionary Template

Column	Data Type	Description	Usage
Ticket ID	Identifier	Unique ticket reference	Record identification
Date	Date/Time	Ticket creation date	Trend analysis
Resolution Date	Date/Time	Ticket resolution date	Resolution calculation
Type	Categorical	Request, Problem or Incident	Ticket classification
Priority	Categorical	Ticket urgency	Priority analysis
Category	Categorical	Technical issue category	Category analysis
Country	Categorical	Ticket country/region	Geographic analysis
Queue / Team	Categorical	Support queue or team	Team performance
Cluster ID	Categorical/Integer	Assigned issue cluster	Cluster analysis
Cluster Name	Categorical	Name of issue cluster	Cluster interpretation
Similarity Score	Numeric	Issue similarity measure	Similarity analysis
Resolution Duration	Numeric	Time from ticket creation to resolution	Performance KPI
Latitude / Longitude	Numeric	Geographic coordinates	Map visualization

End of Report